Dynamics of Möbius Transformations

Fixed points, conjugacy, and iteration on the Riemann sphere

Abstract

A Möbius transformation is a rational map of degree one and a conformal automorphism of the Riemann sphere. Despite its simple formula, its iterates exhibit several distinct behaviors: periodic rotation, quasiperiodic rotation, attraction and repulsion, and parabolic convergence. We derive the complete classification from fixed points and conjugacy, relate it to matrix invariants, and compute explicit examples. All convergence statements are made on the sphere, so that escape to infinity is treated as convergence to an ordinary point of the phase space.

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1 The phase space and the iteration problem

The natural phase space is the Riemann sphere

$$\widehat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}.$$

A *Möbius transformation* is a map

$$T(z) = \frac{az+b}{cz+d}, \qquad a,b,c,d \in \mathbb{C}, \qquad \Delta = ad - bc \neq 0. \tag{1}$$

If $c \neq 0$, its values at the pole and at infinity are

$$T(-d/c) = \infty, \qquad T(\infty) = a/c.$$

If $c = 0$, then $T(\infty) = \infty$. The determinant condition excludes constant maps and ensures that T is bijective, with inverse represented by

$$\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Thus both positive and negative iterates T^n, $n \in \mathbb{Z}$, are defined. Here T^0 is the identity and $T^{n+1} = T \circ T^n$; the notation is composition, not a pointwise power.

To discuss convergence at infinity, use the chordal metric

$$\chi(z, w) = \frac{|z - w|}{\sqrt{1 + |z|^2}\sqrt{1 + |w|^2}}, \qquad \chi(z, \infty) = \frac{1}{\sqrt{1 + |z|^2}}$$

for finite z, w, with $\chi(\infty, \infty) = 0$. In particular, $z_n \to \infty$ on the sphere if and only if $|z_n| \to \infty$.

The basic questions are: which points are fixed or periodic, which orbits converge, and which geometric sets contain the orbits?

2 Matrices, fixed points, and multipliers

2.1 Projective matrix representation

Associate to (1) the matrix

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{GL}(2, \mathbb{C}).$$

In homogeneous coordinates, $z = [z : 1]$ and $\infty = [1 : 0]$, and

$$[u : v] \longmapsto [au + bv : cu + dv].$$

Nonzero scalar multiples of A represent the same transformation. Composition corresponds to matrix multiplication, so A^n represents T^n. Consequently the group of Möbius transformations is $\mathrm{PGL}(2, \mathbb{C})$.

Proposition 2.1. *A nonidentity Möbius transformation has either two distinct fixed points or one fixed point of multiplicity two on $\widehat{\mathbb{C}}$.*

Proof. A projective point is fixed precisely when its representing vector spans an eigenline of A. A nonscalar complex 2×2 matrix has either two distinct eigenlines or one eigenline and a nontrivial Jordan block. Equivalently, the homogeneous fixed-point equation is

$$cu^2 + (d - a)uv - bv^2 = 0.$$

It has two roots on the projective line, counted with multiplicity, unless it vanishes identically. The latter case means $c = b = 0$ and $a = d$, which is exactly the identity transformation. □

For finite fixed points the equation becomes

$$cz^2 + (d - a)z - b = 0. \tag{2}$$

When $c \neq 0$, the two roots are

$$z_\pm = \frac{a - d \pm \sqrt{(a + d)^2 - 4\Delta}}{2c}.$$

When $c = 0$, infinity is fixed and must be included separately; merely solving the finite equation can miss it.

2.2 Multipliers in local coordinates

Definition 2.2. The *multiplier* at a fixed point p is the derivative at zero of $\phi \circ T \circ \phi^{-1}$, where ϕ is a holomorphic local coordinate with $\phi(p) = 0$.

The chain rule shows that this definition does not depend on the chosen coordinate and is invariant under Möbius conjugacy. At a finite fixed point p,

$$T'(p) = \frac{\Delta}{(cp+d)^2}. \tag{3}$$

At infinity, use $u = 1/z$. For example, $T(z) = qz$ has multiplier q at zero and multiplier q^{-1} at infinity, since

$$\frac{1}{T(1/u)} = \frac{u}{q}.$$

A fixed point is called *attracting* if the modulus of its multiplier is less than one, *repelling* if it is greater than one, and *neutral* if it equals one. Multipliers of Möbius maps never vanish.

3 Conjugacy and the two normal forms

Two maps T and S are *Möbius conjugate* if $S = h \circ T \circ h^{-1}$ for a Möbius transformation h. Then

$$T^n = h^{-1} \circ S^n \circ h, \tag{4}$$

so h transports fixed points, periodic points, and orbit limits.

Theorem 3.1 (Normal forms)**.** *Every nonidentity Möbius transformation is conjugate to exactly one of the following types:*

1. $S(w) = qw$*, where* $q \in \mathbb{C} \setminus \{0, 1\}$*, if there are two distinct fixed points;*
2. $S(w) = w + 1$*, if there is one double fixed point.*

In the first case the parameter is determined up to replacement $q \leftrightarrow q^{-1}$*.*

Proof. If the fixed points are α and β, choose h sending them to zero and infinity. When both are finite, one can take

$$h(z) = \frac{z-\alpha}{z-\beta}. \tag{5}$$

When $\beta = \infty$, take $h(z) = z - \alpha$; when $\alpha = \infty$, take $h(z) = 1/(z-\beta)$. The conjugated map fixes zero and infinity, so it is $w \mapsto qw$. Its multiplier at zero is q, which equals the multiplier of T at α. Interchanging the two fixed points inverts q.

If there is only one fixed point p, first send p to infinity. The conjugated map fixes infinity and therefore has the affine form $w \mapsto aw + b$. If $a \neq 1$, it also fixes $b/(1-a)$, a contradiction. Thus it is $w \mapsto w + b$, with $b \neq 0$ because T is not the identity. Rescaling by $w \mapsto w/b$ gives $w \mapsto w + 1$. □

For two distinct finite fixed points, (4) gives an explicit formula for every iterate:

$$T^n(z) = \frac{\beta q^n h(z) - \alpha}{q^n h(z) - 1}. \tag{6}$$

The formula is interpreted on the sphere; a zero denominator represents infinity, and values involving $h(z) = \infty$ are taken by continuity. The reciprocal relation between the two fixed-point multipliers follows immediately from the normal form.

4 The dynamical classification

We use *loxodromic* for the entire class $|q| \neq 1$, and *hyperbolic* for its subclass with q positive real. Some authors reserve "loxodromic" for the nonhyperbolic members of this class. This terminology convention does not change the dynamics.

4.1 Loxodromic maps: attraction and repulsion

Suppose $|q| \neq 1$. Interchange the fixed points if necessary so that $0 < |q| < 1$. In the coordinate $w = h(z)$,

$$S^n(w) = q^n w.$$

Every finite w tends to zero as $n \to +\infty$, while infinity remains fixed. In backward time, every nonzero w tends to infinity.

Proposition 4.1. *A loxodromic map has an attracting fixed point α and a repelling fixed point β. Its forward and backward basins are*

$$\{z : T^n(z) \to \alpha\} = \widehat{\mathbb{C}} \setminus \{\beta\}, \qquad \{z : T^{-n}(z) \to \beta\} = \widehat{\mathbb{C}} \setminus \{\alpha\}.$$

Forward convergence is uniform on compact subsets of $\widehat{\mathbb{C}} \setminus \{\beta\}$, and the analogous statement holds backward. The only periodic points are the two fixed points.

Proof. In the normal coordinate, compact sets avoiding infinity are bounded, and $|q^n w| \leq |q|^n \sup |w| \to 0$ uniformly. Conjugation transports this statement to the sphere. A nonzero finite periodic point would satisfy $q^n w = w$ for some positive n, requiring $q^n = 1$, which is impossible when $|q| \neq 1$. □

Writing $q = \rho e^{i\theta}$, a nonzero orbit satisfies

$$|w_n| = \rho^n |w_0|, \qquad \arg w_n \equiv \arg w_0 + n\theta \pmod{2\pi}.$$

It combines dilation with rotation. If $\theta \neq 0$, the orbit lies on the continuous logarithmic spiral $w(t) = w_0 \exp(t(\log \rho + i\theta))$, sampled at integer times. For $q > 0$, there is no rotation: each ray from zero is preserved. This is the hyperbolic case. A negative real q instead produces a half-turn at every step and is not hyperbolic under our convention.

4.2 Elliptic maps: periodic and irrational rotations

Suppose $|q| = 1$ and $q \neq 1$. Write $q = e^{i\theta}$. The map is called *elliptic*. Both fixed points are neutral, and every circle $|w| = r$, $0 < r < \infty$, is invariant.

Proposition 4.2. *There are two possibilities for an elliptic map.*

1. *If $\theta/(2\pi) = p/m \in \mathbb{Q}$ in lowest terms, then T^m is the identity. Every point other than the two fixed points has exact period m.*

2. *If $\theta/(2\pi) \notin \mathbb{Q}$, there are no nonfixed periodic points. Each other orbit is dense in the generalized circle $h^{-1}(\{|w| = |h(z)|\})$.*

Proof. A nonzero finite point is periodic exactly when $q^n = 1$ for some $n > 0$. This yields the rational case and excludes additional periodic points in the irrational case. For irrational $\theta/(2\pi)$, the subgroup $\{e^{in\theta} : n \in \mathbb{Z}\}$ is dense in the unit circle: its closure is an infinite closed subgroup of the circle, and every proper closed subgroup of the circle is finite. Forward powers have the same closure. Multiplying by w_0 and applying h^{-1} proves the orbit-density claim. □

Here a *generalized circle* means a Euclidean circle or a Euclidean line together with infinity. Möbius transformations send generalized circles to generalized circles. Irrational elliptic orbits do not converge to a fixed point; their entire invariant circle is their accumulation set.

4.3 Parabolic maps: one neutral fixed point

A nonidentity map with a double fixed point is called *parabolic*. In its normal coordinate,

$$S(w) = w + 1, \qquad S^n(w) = w + n.$$

Its unique fixed point is infinity, and its multiplier there is one:

$$u \longmapsto \frac{u}{1+u}, \qquad \frac{d}{du}\frac{u}{1+u}\bigg|_{u=0} = 1.$$

Proposition 4.3. *If T is parabolic with fixed point p, then*

$$T^n(z) \longrightarrow p \quad \text{as } n \to +\infty \quad \text{and as } n \to -\infty$$

for every $z \in \widehat{\mathbb{C}}$. The only periodic point is p. Convergence is uniform on compact subsets of $\widehat{\mathbb{C}} \setminus \{p\}$, but not on the whole sphere.

Proof. For finite w, $|w + n| \to \infty$ in either direction, and this convergence is uniform for bounded sets of w. No finite point satisfies $w + n = w$ for a nonzero integer n. Uniform convergence on the entire sphere is impossible: for each positive n, the initial point $w = -n$ maps to zero, whose chordal distance from infinity is one. □

Remark 4.4 (Convergence is not the same as attraction). Although all parabolic orbits converge to p, the fixed point is neutral, not attracting in the multiplier sense. It is not even Lyapunov stable: for $S(w) = w + 1$, initial points $w = -N$ can be arbitrarily close to infinity, yet $S^N(-N) = 0$. Thus pointwise convergence does not imply that nearby orbits remain nearby at every intermediate time.

Horizontal lines $\operatorname{Im} w = t$, completed by infinity, are invariant under translation. Their inverse images under h are invariant generalized circles through the parabolic fixed point. Distinct members of this family meet only at that point and are tangent there.

4.4 The identity and a summary

The identity must be separated from the parabolic class: it fixes every point, and every orbit is constant. In particular, multiplier one alone does not imply a nontrivial parabolic transformation.

Type	Normal form	Nonfixed forward orbits
Identity	w	None
Parabolic	$w + 1$	Converge to the double fixed point
Elliptic, rational angle	$e^{2\pi ip/m}w$	Periodic of period m
Elliptic, irrational angle	$e^{i\theta}w$	Dense on invariant circles
Loxodromic	qw, $0 < \lvert q\rvert < 1$	Converge to the attracting point

In the last row, the repelling fixed point itself remains fixed.

5 Classification directly from matrix data

The quantity

$$\tau(T) = \frac{(\operatorname{tr} A)^2}{\det A} \tag{7}$$

is unchanged by rescaling A and by matrix conjugation. If A has distinct eigenvalues λ_1, λ_2, the normal-form parameter is their ratio in one order or the other, and

$$\tau = q + q^{-1} + 2. \tag{8}$$

More specifically, the multiplier at the eigenline for λ_1 is λ_2/λ_1: in an eigenvector basis the transverse coordinate is multiplied by this ratio.

Corollary 5.1 (Trace criterion)**.** *For a nonidentity Möbius transformation:*

1. $\tau = 4$ *if and only if the map is parabolic;*
2. $\tau \in [0, 4) \subset \mathbb{R}$ *if and only if the map is elliptic;*
3. $\tau \notin [0, 4] \subset \mathbb{R}$ *if and only if the map is loxodromic;*
4. $\tau \in (4, \infty) \subset \mathbb{R}$ *if and only if the map is hyperbolic.*

The identity also has $\tau = 4$, so it must be checked separately.

Proof. The repeated-eigenvalue condition is $(\operatorname{tr} A)^2 = 4 \det A$. A nonscalar matrix with this property gives the parabolic normal form. For $q = e^{i\theta} \neq 1$, (8) gives $\tau = 2 + 2\cos\theta \in [0, 4)$. Conversely, if $\tau \in [0, 4)$, the roots of $q^2 - (\tau - 2)q + 1 = 0$ lie on the unit circle and are not one. The remaining nonparabolic cases have $|q| \neq 1$. Finally $q > 0$, $q \neq 1$, is equivalent to $q + q^{-1} > 2$, or $\tau > 4$. □

Remark 5.2. One may normalize A to determinant one, in which case $\tau = (\operatorname{tr} A)^2$. For complex matrices, however, the shortcut of comparing $|\operatorname{tr} A|$ with two does not classify the dynamics. For example, $A = \operatorname{diag}(i/\sqrt{2}, -i\sqrt{2})$ has determinant one and $|\operatorname{tr} A| = 1/\sqrt{2} < 2$, but its eigenvalue ratio is $-1/2$; the transformation is loxodromic, not elliptic.

6 Worked examples

Example 6.1 (A hyperbolic map with finite fixed points)**.** Consider

$$T(z) = \frac{2z + 1}{z + 2}.$$

Its fixed-point equation is $z^2 - 1 = 0$, so the fixed points are 1 and -1. Since $\Delta = 3$,

$$T'(1) = \frac{1}{3}, \qquad T'(-1) = 3.$$

For $h(z) = (z - 1)/(z + 1)$ a direct calculation gives $h(T(z)) = h(z)/3$. Consequently

$$T^n(z) = \frac{1 + 3^{-n} h(z)}{1 - 3^{-n} h(z)}.$$

Every point except -1 converges to 1 forward in time. The matrix $\left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ gives $\tau = 16/3 > 4$, confirming the hyperbolic classification.

Example 6.2 (A genuinely rotating loxodromic map)**.** Let $T(z) = (i/2)z$. Zero is attracting, infinity is repelling, and

$$T^n(z) = 2^{-n} i^n z.$$

Each step rotates through $\pi/2$ while halving the radius. Every finite orbit tends to zero. The iterates visit four rays in succession, showing that the rotational part can be periodic even when the full orbit is not.

Example 6.3 (Finite-order elliptic dynamics)**.** Let

$$T(z) = \frac{z-1}{z+1}.$$

The fixed points are i and $-i$. At i the multiplier is $2/(1+i)^2 = -i$, a primitive fourth root of unity. Direct iteration gives

$$T^2(z) = -\frac{1}{z}, \qquad T^4(z) = z.$$

Thus every nonfixed point has exact period four; for instance,

$$0 \longmapsto -1 \longmapsto \infty \longmapsto 1 \longmapsto 0.$$

Here $\tau = 2$, in agreement with the elliptic trace criterion.

Example 6.4 (An irrational elliptic orbit)**.** Let $T(z) = e^{2\pi i\sqrt{2}} z$. Zero and infinity are fixed, and every other orbit is dense on the circle $|z| = |z_0|$. No other point is periodic, because $n\sqrt{2} \notin \mathbb{Z}$ for every positive integer n. The orbit stays bounded in the finite plane but does not converge.

Example 6.5 (A parabolic map with a finite fixed point)**.** Consider

$$T(z) = \frac{z}{z+1}.$$

Its only fixed point is zero, of multiplicity two. The coordinate $h(z) = 1/z$ gives $h(T(z)) = h(z) + 1$, hence

$$T^n(z) = \frac{z}{1+nz}, \qquad n \in \mathbb{Z}. \tag{9}$$

For fixed finite $z \neq 0$, this tends to zero at the algebraic rate $T^n(z) \sim 1/n$, rather than the exponential rate of a loxodromic map. Also $T^n(\infty) = 1/n$ for $n \neq 0$. A pole at an intermediate iterate does not terminate the orbit: it merely sends that iterate to infinity. Finally $T'(0) = 1$ and $\tau = 4$, so this is parabolic, not attracting in the multiplier sense.

7 A practical procedure and final perspective

Given coefficients a, b, c, d, the following procedure determines the dynamics without computing a long sequence of iterates.

1. Verify $ad - bc \neq 0$. If $b = c = 0$ and $a = d$, the map is the identity.
2. Solve the homogeneous fixed-point equation, remembering infinity.
3. With two distinct fixed points, send them to zero and infinity; compute q as the multiplier at the point sent to zero.

4. If $|q| \neq 1$, identify the attracting and repelling points. If $|q| = 1$, decide whether q is a root of unity.

5. With one double fixed point, conjugate to translation.

6. Use $T^n = h^{-1} \circ S^n \circ h$ for explicit orbits and limits. The invariant τ provides an independent classification check.

The central observation is that degree-one dynamics are completely governed by a dilation–rotation or a translation. There are no critical points, no coexistence of distinct attracting cycles, and no nontrivial periodic cycles except in the finite-order elliptic case. Nevertheless, the examples separate several concepts that are easy to confuse: neutral and attracting fixed points, bounded and convergent orbits, pointwise and uniform convergence, and Euclidean escape versus convergence on the sphere.